

Selection, agreement, and generation

A mechanism comparison of AutoGaze, DOUBT, and DiTFuse

Visual models can change what they process, estimate whether their answers are trustworthy, or learn how to synthesize an output. AutoGaze, DOUBT, and DiTFuse instantiate these distinct interventions. Their common use of repeated model calls conceals different state variables and control signals: a selected-patch prefix, two sampled-answer sets, and an evolving target latent, respectively. The distinction matters because a reconstruction-loss prediction, response concentration, and a learned velocity field support different inferences about the resulting system. This comparison describes mechanisms; it reports no new model execution, accuracy measurement, or benchmark replication.

	AutoGaze	DOUBT	DiTFuse
Intervention	Select visual evidence before ViT	Score sampled responses	Generate a task-conditioned image
Repeated state	Patch-index history	Separate response sets	Noisy target latent
Inference control	Predicted loss versus ϵ	Mean resultant versus θ	Iterative velocity-based updates
Learning role	Train gaze policy and loss head	Frozen detector components	Train LoRA adapters

1 AutoGaze: learn what to observe

AutoGaze [1] selects multi-scale patches before the expensive downstream vision transformer (ViT). A video is divided into $16 \times 224 \times 224$ spatiotemporal tiles. Within each tile, a lightweight convolutional encoder processes frames sequentially. An autoregressive transformer uses current and past frame features together with the selected-index prefix to predict a patch index and the reconstruction loss expected after that selection. Appending the selected index makes subsequent choices depend on the existing gaze history (Figure 1, panel A).

The vocabulary maps indices to scale and location at image resolutions 32, 64, 112, and 224 pixels; these values are not patch side lengths. Gazing stops for the current frame when the *predicted* loss is strictly below the user threshold ϵ , then advances to the next frame. Otherwise selection continues within that frame. This follows the accepted paper’s main-text comparator, rather than the illustrative non-strict inequality in its original Figure 3. The criterion is not a guarantee that actual reconstruction error meets the threshold. Optional multiple prediction heads retain the history dependence.

Selected tile positions are merged into original-video coordinates. Scale-specific patch embeddings and interpolated positional embeddings yield selected tokens from all 16 frames for a downstream ViT sequence and then the MLLM. Thus selection changes the evidence entering attention, rather than pruning an already computed downstream attention map. Training supplies the missing reconstruction signal: a custom block-causal VideoMAE measures weighted pixel and perceptual loss. Greedy-search sequences first supervise index prediction and squared loss-prediction error; on-policy GRPO subsequently uses negative actual reconstruction loss as reward, with loss-head supervision at each frame’s last selected patch.

Figure 1: AutoGaze: sequential selection before downstream attention (facing page). Panel A separates causal frame/index history, patch gathering, and the predicted-loss stopping decision. A yes advances the frame; a no continues gazing. Panel B merges tile positions and embeds selected multi-scale patches before the ViT and MLLM. Panel C isolates training: measured reconstruction loss supplies greedy-search targets, GRPO rewards, and loss-head supervision; it is not recomputed by VideoMAE to make the depicted inference stop decision. Solid, dashed, and dotted routes distinguish data/index feedback, stop control, and training relations. Based on the CVPR 2026 accepted version, Sections 3.1–3.3 and Figure 3 [1].

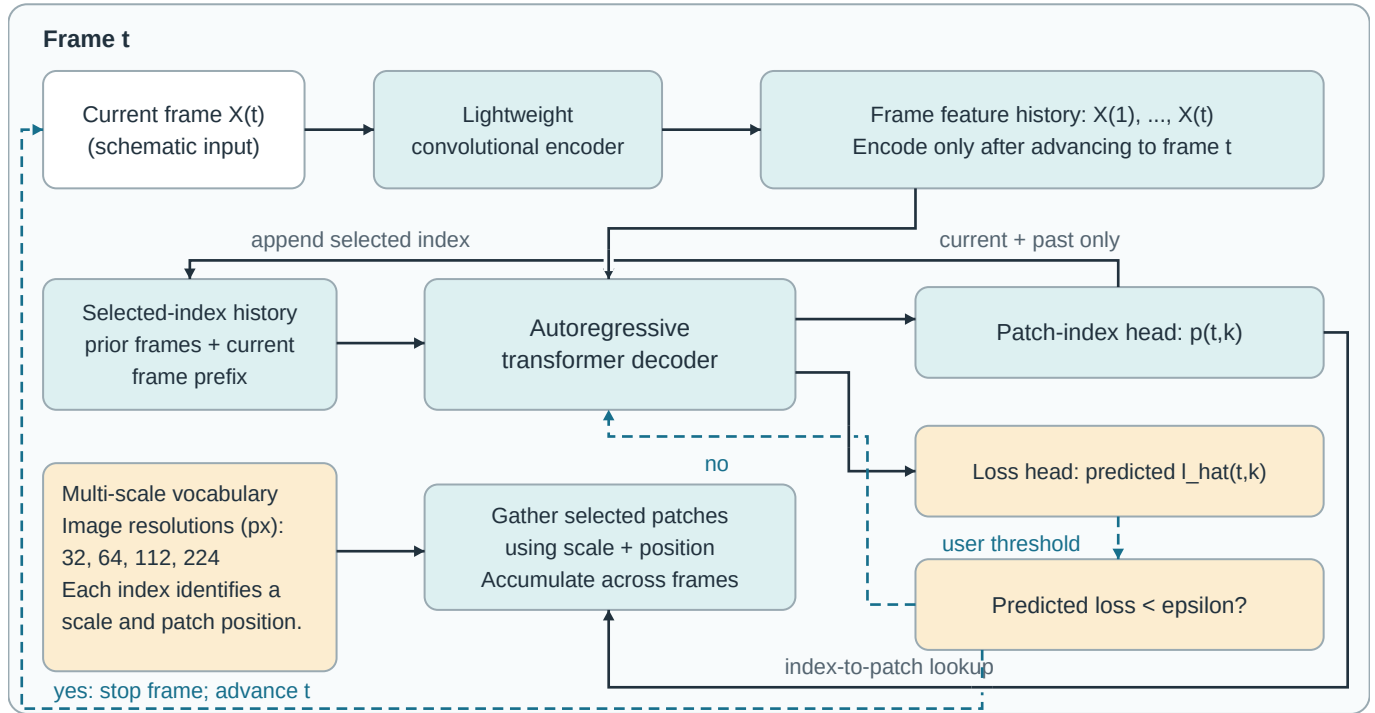
AutoGaze Select multi-scale patches before the downstream ViT

A Inference: frame-by-frame gazing within each tile

Any-length / resolution video

Tile into 16 x 224 x 224 blocks

Repeat panel A per tile



Optional multi-token prediction

The autoregressive history is retained.

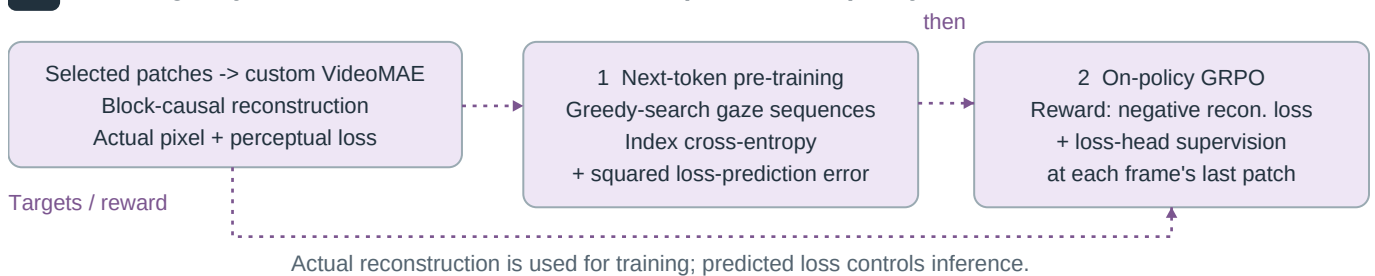
Several heads emit multiple indices and matching predicted losses per decoder step.

B Downstream: merge positions, embed selected patches, then attend

Panel A outputs from all tiles



C Training only: measured reconstruction loss supervises the policy



→ data / index feedback

- - - - - stop / continue control

- - - - - training supervision

Original mechanism schematic; not an empirical gaze example. Source: Shi et al., CVPR 2026, Fig. 3 and Sec. 3.

2 DOUBT: concentration as a detection proxy

DOUBT [2] interrogates one black-box MLLM M through two prompt conditions. The direct condition collects K responses to image I and instruction t . In the second condition, the same model recognizes up to three main objects, producing $r = M(I, p_{\text{obj}})$; bridging then collects K responses to $M(I, t \| p_{\text{bri}} \| r)$, where $\|$ means prompt concatenation. The bridge retains both the image and original instruction. It does not replace them with an object list.

A shared frozen `nli-roberta-large` encoder embeds each response. For set $s \in \{\text{ori}, \text{bri}\}$, individual normalization precedes the within-set reduction:

$$\hat{e}_{s,k} = \frac{f(a_{s,k})}{\|f(a_{s,k})\|_2}, \quad R_s = \frac{1}{K} \left\| \sum_{k=1}^K \hat{e}_{s,k} \right\|_2, \quad R_{\text{avg}} = \frac{R_{\text{ori}} + R_{\text{bri}}}{2}. \quad (1)$$

Here $a_{s,k}$ denotes the k th response in either set. The two scalar resultant lengths are averaged *after* separate reductions. Pooling the $2K$ vectors before taking a norm would define a different score. The rule returns flag 1 (non-hallucinatory) when $R_{\text{avg}} > \theta$ and flag 0 (hallucination) when $R_{\text{avg}} \leq \theta$. The equality case therefore belongs to flag 0.

Response concentration is a proxy for trustworthiness, not a verification of image-grounded truth. In particular, high concentration within each set does not require the direct and bridged sets to agree with one another. For an illustrative algebraic limiting case, if every direct embedding is the same unit vector u and every bridged embedding is $-u$, both resultant lengths equal 1 and so does R_{avg} . This is a consequence of Equation 1, not an observed model outcome. Consistently incorrect answers can therefore evade a disagreement-based warning, and an erroneous object list can propagate through bridging. Neither independence of the two scores nor unbiasedness of the finite-sample estimator is established here.

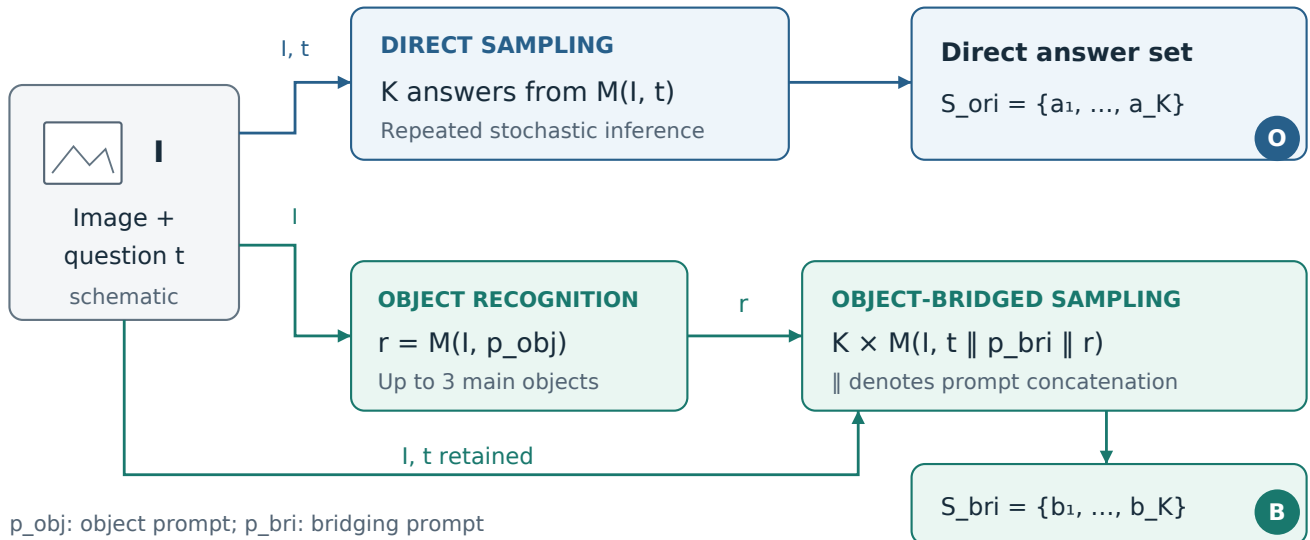
The low-temperature reference answer A is compared with ground truth only for offline evaluation. Neither A nor ground truth enters Equation 1. The reported choices $K = 10$, reference/sample temperatures 0.1/0.5, top- $k = 10$, top- $p = 0.99$, and $\theta = 0.48$ are experimental settings, not architectural constants. The DOUBT account here follows the supplied grounded method description of Sections 3.1–3.3, Equations 4–6, and Algorithm 1; the original paper was not independently inspected for this comparison.

The contrast with AutoGaze is structural. AutoGaze’s scalar prediction governs how much additional visual evidence is selected. DOUBT’s scalar statistic classifies the trustworthiness of sampled responses after model inference. Its repeated calls do not update a gaze policy or detector weights. A threshold crossing therefore has a different meaning in the two systems: terminating evidence acquisition versus issuing a detection estimate.

Figure 2: DOUBT: object-bridged sampling and separate-set scoring (facing page). Panel A preserves the same MLLM and the original image/instruction in both sampling paths. Panel B shares a frozen encoder but keeps direct and bridged vectors separate through normalization, summation, norm, and division by K ; only the scalar results are averaged. High R_{avg} yields flag 1, with equality assigned to flag 0. Panel C isolates offline reference-answer evaluation: its dashed data routes do not feed detection. O/B labels continue the same answer sets across panels. The schematic follows the supplied account of the PMLR 306 (2026) version [2]; concentration is a detection proxy and object-recognition errors can propagate.

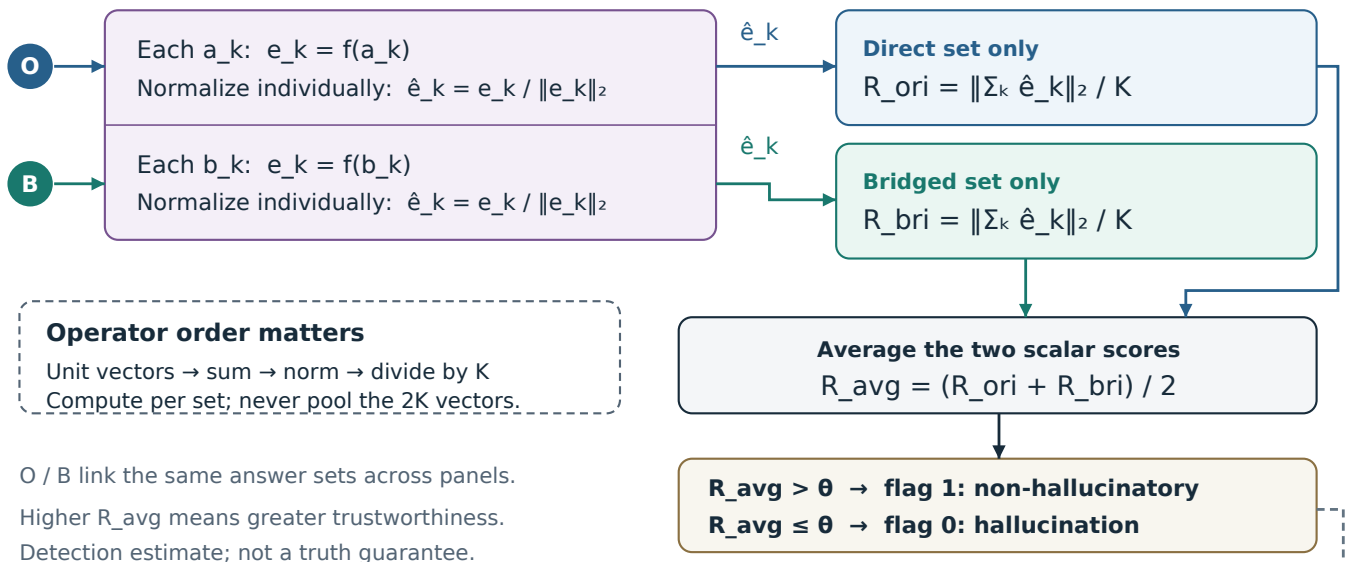
DOUBT Object-bridged sampling → separate-set trustworthiness

A Repeated inference with one black-box MLLM M



B Frozen embeddings, then two separate resultant lengths

Same frozen nli-roberta-large encoder f



C Offline evaluation only — reference A never enters the score



3 DiTFuse: adapt a conditional velocity field

DiTFuse [3] generates a task-consistent image through one shared conditional diffusion transformer (Figure 3, panel A). The two input images remain distinct through a frozen SDXL VAE and visual patch embedding with patch size 2. Phi-3 text embeddings represent the task-tagged instruction. Interleaved text and delimited image tokens precede the timestep and noisy-target tokens in a 32-layer pretrained OmniGen/Phi-3 backbone. Its hybrid mask permits causal dependencies across blocks and bidirectional attention within each image block. Target queries can read earlier conditions and time, plus their own target block; earlier condition blocks cannot read future target tokens. Only target-token states are reconstructed into the latent velocity.

Training selects one instruction-appropriate target image and encodes it to latent x . M3 constructs two degraded source copies using aligned patch grids with approximately 75% complementary and 25% common corruption, and uses the original source as target. Other targets depend on the task: brightness/contrast adjustments, a translucent blue target-class overlay, or fusion supervision. With Gaussian noise ε and sampled $t \in [0, 1]$, flow matching forms

$$x_t = tx + (1 - t)\varepsilon, \quad \mathcal{L} = \mathbb{E} \|(x - \varepsilon) - v_\theta(x_t, t, c)\|_2^2.$$

The target-selection bus chooses one target; it does not sum the candidate targets. This constructed supervision precludes describing the whole method as unsupervised.

The depicted author-paper adaptation places rank-64 LoRA on all linear layers with scaling 0.5 while freezing the backbone. The supplied code description instead targets `qkv_proj/o_proj` and sets `lora_alpha` equal to rank. These settings are not asserted to be equivalent; the diagram follows the paper, and the code was not independently inspected. At inference the adapted weights stay fixed: sampling starts from Gaussian noise, repeatedly updates the target latent using the predicted velocity (50 steps in the source schematic), and decodes the final latent. No target image, training loss, or parameter update enters that loop.

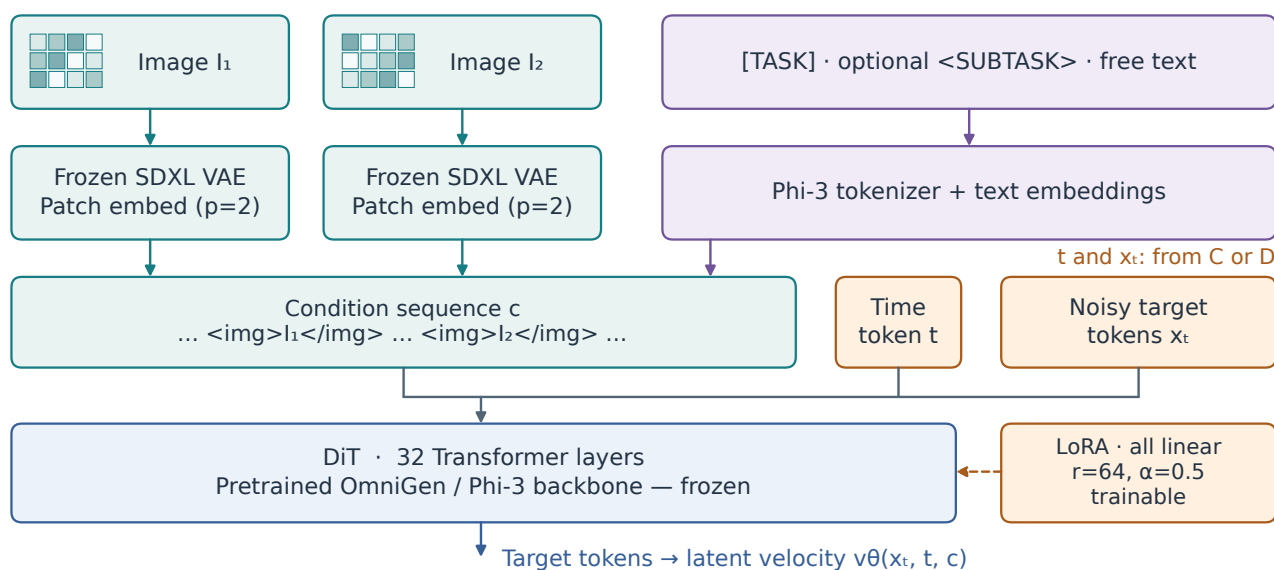
Across the three methods, the object being controlled explains the useful comparison: AutoGaze controls visual evidence, DOUBT estimates answer trustworthiness, and DiTFuse controls generated image content. Their loops are not interchangeable optimization procedures. Likewise, stopping, detection, and conditional generation do not provide comparable quality guarantees. The architectures motivate these distinctions but supply no common-task evidence for ranking their accuracy or efficiency.

Figure 3: DiTFuse: one shared model for training and generation (facing page). A shows the paper’s frozen backbone with LoRA; B shows allowed attention dependencies, not measured scores. C separates task-target selection from latent flow supervision. The fusion target marked with an asterisk is mean fusion for infrared–visible fusion (IVIF), a normally exposed target in synthetic multi-exposure fusion (MEF), or provided pseudo ground truth for multi-focus fusion (MFF). D reuses A with fixed adapted parameters. Solid arrows carry data/conditioning, dashed arrows denote parameter adaptation or training updates, and the return loop denotes repeated inference. Token counts are schematic. Based on arXiv:2512.07170v1, Sections III-A/B, Figures 3, 4, and 6, and Equations 4–5 [3]; the paper/code LoRA mismatch is stated above.

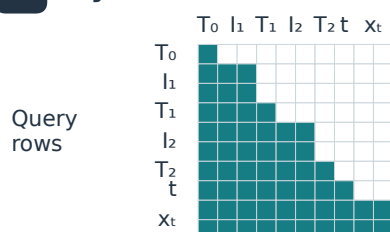
References

- [1] B. Shi et al. *Attend Before Attention: Efficient and Scalable Video Understanding via Autoregressive Gazing*. CVPR 2026, pp. 17022–17034. CVP accepted-paper open-access version.
- [2] *DOUBT: Decoupled Object-level Understanding and Bridging via vMF-based Trustworthiness for Hallucination Detection in MLLMs*. PMLR 306 (2026), pp. 16418–16441. <https://proceedings.mlr.press/v306/chen26d1.html>.
- [3] Li et al. *Towards Unified Semantic and Controllable Image Fusion: A Diffusion Transformer Approach*. arXiv:2512.07170v1, 8 December 2025, author manuscript; associated DOI: 10.1109/TPAMI.2025.3642842.

Schematic inputs



Allowed dependencies, not scores



Key columns →

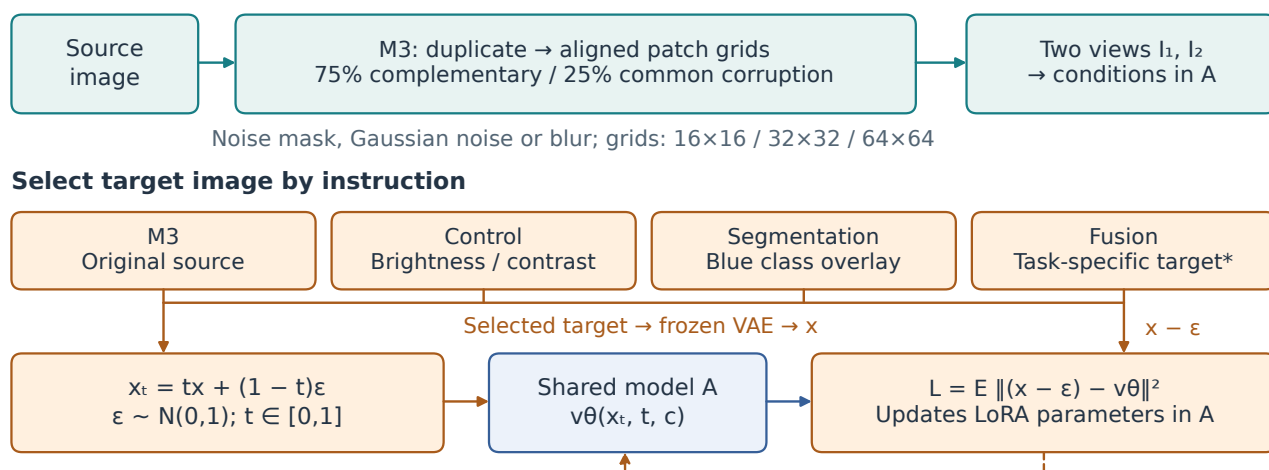
Across blocks: attend to earlier tokens only.

Inside l_1 , l_2 and x_t : bidirectional attention.

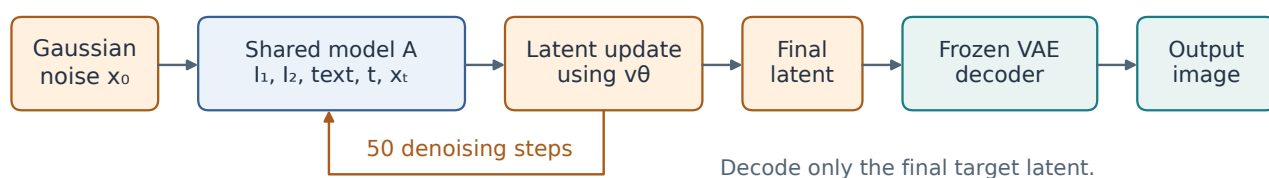
Target x_t sees all conditions, t and its own block.

$T_0 / T_1 / T_2 =$ interleaved text

Task-aware data + one flow-matching objective



Fixed adapted weights; no loss or target image



→ Data / conditioning

Parameter adaptation / training update (A, C)

↑ Inference repeat (D)